The elimination method

1 Consider the following linear equations:

$$3x + 7y = -6$$
$$2x - y = -\frac{45}{7}$$

a What number should the second equation be multiplied by to eliminate y?

b Eliminate y and hence find the value of x that satisfies both equations.

c Substitute the x-value to find the value of y that satisfies both equations.

2 Solve the following systems of equations:

a $3x + 6y = 2$
$3x + 3y = -7$

b $8x + 3y = -11$
$-8x - 5y = 29$

c $3x - 5y = 57$
$3x + 7y = -51$

d $8x + 3y = 19$
$3x - 3y = 25$

e $8x + 3y = -9$
$5x + 3y = 3$

f $3x + 4y = 12$
$-3x + 6y = 18$

g $7x + 9y = -41$
$5x - 18y = 6$

h $6x - 21y = -21$
$8x + 7y = 27$

i $8x + 7y = -30$
$6x + 28y = -68$

j $7x + 8y = \frac{4}{3}$
$4x + 3y = -27$

k $3x + y = 4$
$2x - y = 11$

l $11x + 7y = -6$
$2x - y = -17$

m $-8x - y = 5$
$-6x + 3y = 5$

n $7x + 11y = -65$
$-21x + 7y = 75$

o $9x + 11y = -29$
$27x + 8y = -37$

p $10x + 6y = -8$
$5x + 10y = -25$

q $2x + 3y = 4$
$x - 2y = 9$

r $4x - 9y = 3$
$-5x + 7y = -8$

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3 For each of the following systems of equations:

i Rewrite the system of equations as an equivalent system with integer coefficients, keeping the integers as small as possible.

ii Solve for x and y.

a $3x + 11y = -48$
$-2x + 7y = -\frac{76}{3}$

b $0.2x + 0.3y = 0.5$
$0.5x + 0.4y = 0.2$

c $\frac{4x}{5} + \frac{3y}{5} = 4$
$8x - 3y = 4$

d $\frac{x}{5} + \frac{y}{6} = 8$
$\frac{x}{6} + \frac{y}{2} = 11$

4 Use the elimination method to solve the following pairs of equations:

a $2x + 5y = 44$
$6x - 5y = -28$

b $8x + 3y = -11$
$-8x - 5y = 29$

c $2x - 5y = 1$
$-3x - 5y = -39$

d $7x - 4y = 15$
$7x + 5y = 60$

e $-6x - 2y = 46$
$-30x - 6y = 246$

f $-5x + 16y = 82$
$25x - 4y = 122$

g $-\frac{x}{4} + \frac{y}{5} = 8$
$\frac{x}{5} + \frac{y}{3} = 1$

h $\frac{4x}{5} + \frac{3y}{5} = 7$
$8x - 3y = 1$

i $0.4x - 0.63y = 0.23$
$2x + 7y = -9$

j $5x + 3y = 7$
$x + y = 2$

k $-5p - 7q = -\frac{43}{5}$
$-18p - 28q = -\frac{187}{5}$

l $0.2x + 0.3y = 0.5$
$0.5x + 0.4y = 0.2$

Applications

5 When comparing some test results, Eileen noticed that the sum of her Geography test score and Maths test score was 122. She also noticed that their difference was 12. She knows that she scored higher on the Geography test. Given that her Geography score is x and her Maths score is y:

a Use the sum of the test scores to set up Equation 1.

b Use the difference of the test scores to set up Equation 2.

c Solve for x to find Eileen's Geography score.

d Hence, solve for y to find Eileen's Maths score.

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6 9 pens and 2 rulers cost $\$26$ while 3 pens and 4 rulers cost $\$22$. Let x and y be the price of the pen and ruler respectively.

a Use the fact that 9 pens and 2 rulers cost $\$26$ to set up Equation 1.

b Use the fact that 3 pens and 4 rulers cost $\$22$ to set up Equation 2.

c Solve for x to find the price of each pen.

d Hence, solve for y to find the price of each ruler.

7 The proportion of the workforce of a particular country that are teenagers is modelled by $3.3x + y = 70.8$, where x is the number of years since 2015 and y is the proportion as a percentage. The proportion of the workforce that are pensioners is modelled by $3.2x - y = -64.5$.

a Use the elimination method to solve for x. Give your answer to the nearest whole number.

b Use this rounded value of x to solve for y. Give your answer to the nearest whole number.

c Estimate the year in which the proportion of the workforce that are teenagers and the proportion of the workforce that are pensioners are the same.

d Hence, estimate the percentage of the workforce that are teenagers (or the percentage of the workforce that are pensioners) during the estimated year in part (c).

8 For two numbers, x and y:

- Two times the x is added to y to get 27.
- The difference between five times x and y is 43.

a Set up two equations for the given information.

b Solve for x and y.

9 For two numbers, x and y:

- The sum of the x and y is 10.
- The difference between six times x and five times y is 5.

a Set up two equations for the given information.

b Solve for x and y.

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10 For two numbers, x and y:

- Seven times x is added to y and is equal to 64.
- The difference between three times x and y is 16.

a Set up two equations for the given information.

b Solve for x and y.

11 A mother is currently 10 times older than her son. In 3 years time, she will be 7 times older than her son. Let x and y be the present ages of the son and mother respectively.

a Write two equations for the given information.

b Find the current ages of the mother and son.

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